

Newton's First Law

Grade: 8

Subject: Physics (HKUST Press Edition)



Question What makes objects move, and what makes them tend to stop?



① Water droplet in free fall



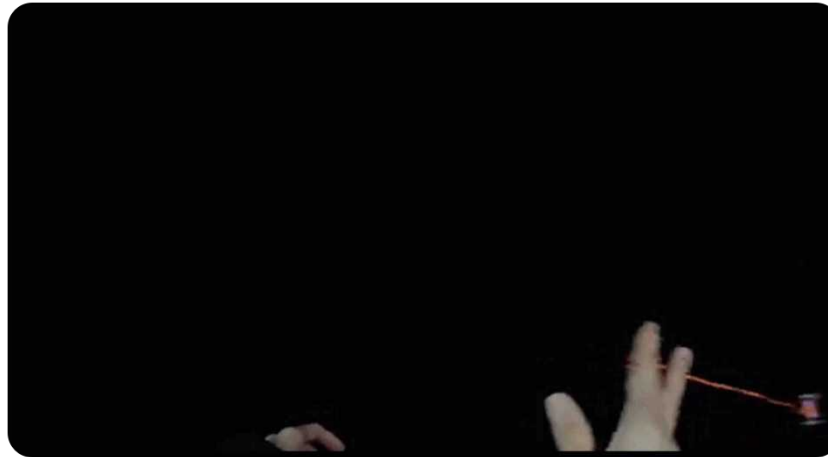
② Planet orbiting the Sun



③ Skateboard rolling down a ramp



④ Suitcase rolling forward



⑤ Yo-yo swung in a circle

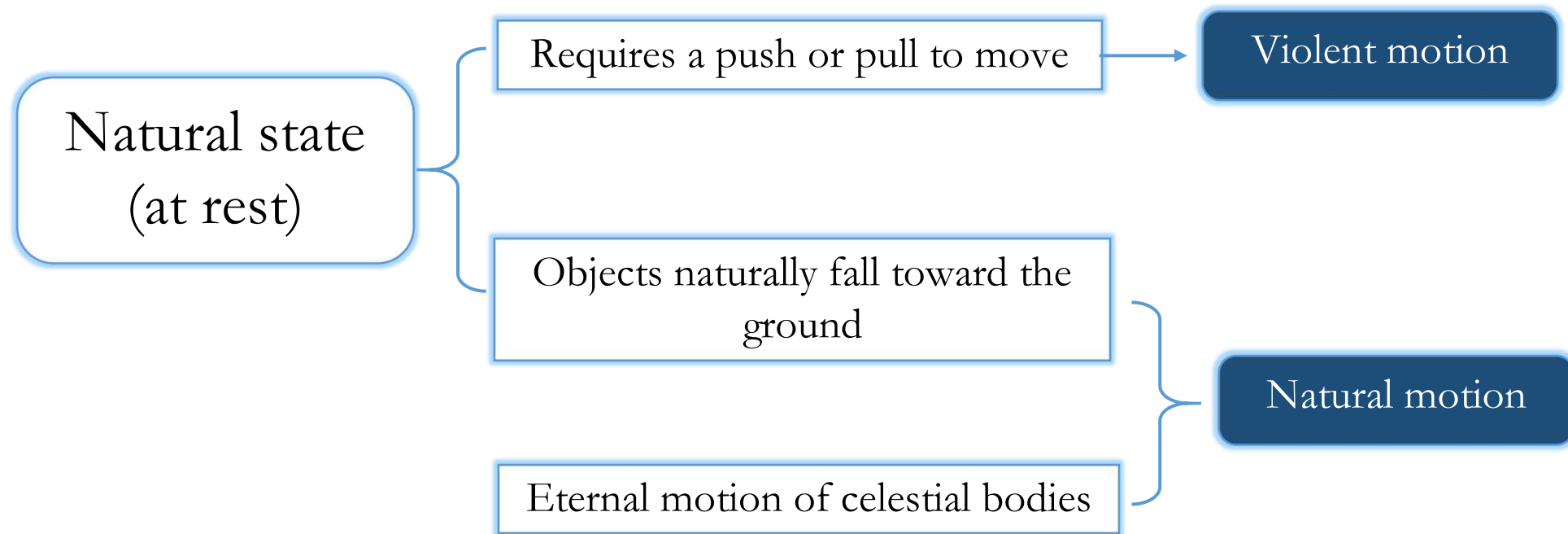


⑥ Person pulling a suitcase forward



Question

What makes objects move, and what makes them tend to stop?





Question What makes objects move, and what makes them tend to stop?

Viewpoint1 Aristotle, fourth century BCE

Natural state
(at rest)

Requires a push or pull
to move

Only continuous force can keep an
object in motion.

Force is required to maintain motion.

Objects naturally fall
toward the ground

The Earth is the center of the cosmos;
objects naturally move toward the center
without any evident external force.

Eternal motion of
celestial bodies

Celestial matter has special intrinsic
properties that allow it to sustain its own
motion.

Question

Is a force required to maintain motion?

Requires a push or pull to move

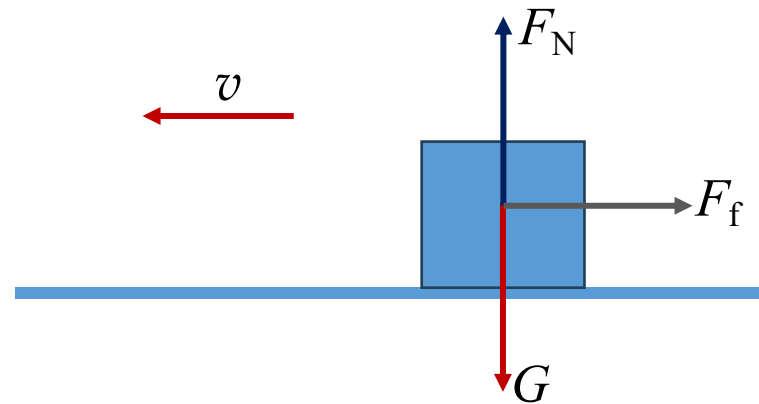
Violent motion



Suitcase rolling
forward

Force is required to maintain motion

Observation + induction

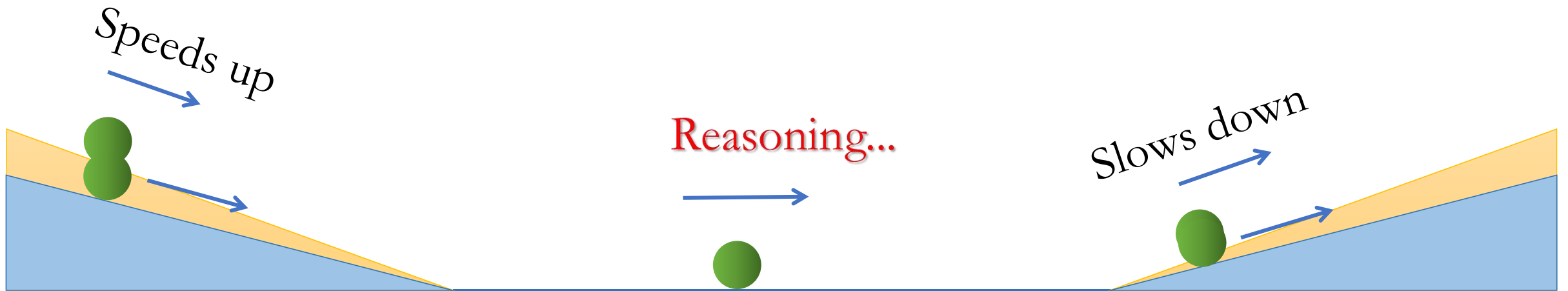


Objects naturally come to rest

Natural motion

Question

Is a force required to maintain motion?

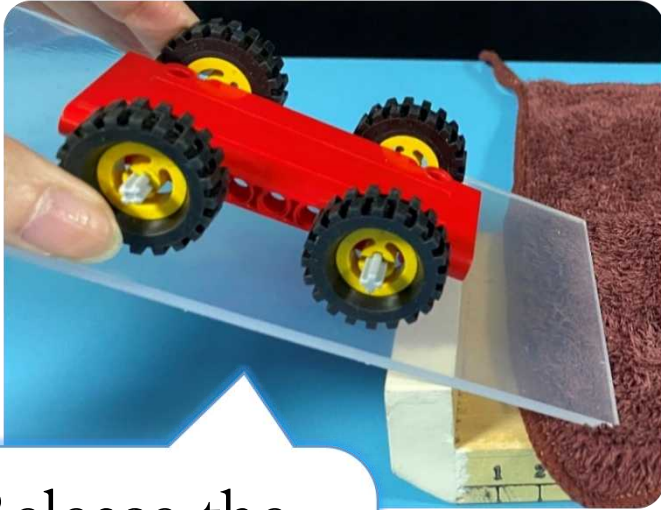


Galileo
(early seventeenth century)

An object moving on a horizontal plane
maintains a constant speed.

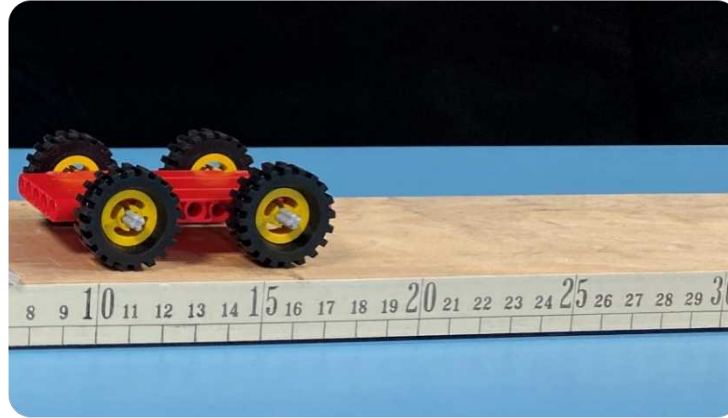
Task: Design an experiment to demonstrate that a force is not required to maintain motion.

v



Release the cart from the same height

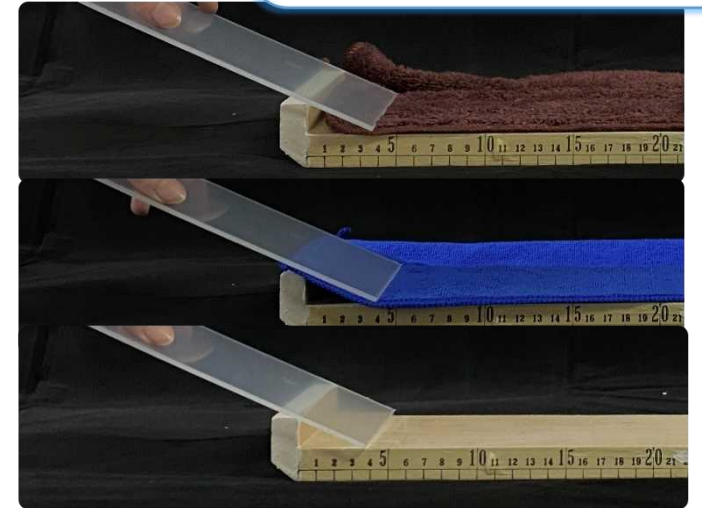
Discussion:
Experimental plan



Galileo
(early seventeenth century)

F_f

Reduce surface roughness



An object moving on a horizontal plane maintains a constant speed.